>>> 50 MongoDB Question

Introduction to MongoDB:

1. What is MongoDB, and how does it differ from traditional relational databases?
2. Explain the concept of NoSQL databases and where MongoDB fits in.
3. What are the key features of MongoDB that make it popular for modern applications?
4. Describe the JSON-like structure of MongoDB documents.
5. How does MongoDB handle data relationships compared to relational databases?

Basic MongoDB Operations:

6. What are the basic operations for inserting documents into a MongoDB collection?
7. How do you retrieve documents from a MongoDB collection?
8. Explain how to update documents in MongoDB using the `update` method.
9. Describe the purpose of the `find()` method in MongoDB.
10. How do you delete documents from a MongoDB collection?

Indexes and Query Performance:

11. What is an index in MongoDB, and why is it important for query performance?
12. How do you create an index on a field in a MongoDB collection?
13. Explain the different types of indexes in MongoDB (e.g., single, compound, text, and geospatial).
14. How can you check which indexes are available on a MongoDB collection?
15. What is the significance of the "explain" method in MongoDB?

MongoDB Data Modeling:

16. How do you model relationships between documents in MongoDB?
17. What are embedded documents, and when should you use them in MongoDB?
18. Explain the concept of database references (DBRefs) in MongoDB.
19. How do you design a schema to avoid excessive nesting in MongoDB?
20. What is denormalization, and when is it appropriate in MongoDB?

Aggregation Framework:

21. What is the MongoDB Aggregation Framework, and how does it work?
22. Explain the purpose of aggregation stages like `$match`, `$group`, and `$sort`.
23. How can you project specific fields in MongoDB aggregation queries?
24. What are pipeline operators, and how are they used in aggregation queries?
25. Describe how to use the `$lookup` stage for performing joins in MongoDB.

Replication and High Availability:

26. What is replication in MongoDB, and why is it important?

27. Explain the role of primary and secondary nodes in a MongoDB replica set.
28. How do you set up and configure a MongoDB replica set?
29. What is the purpose of an "oplog" in MongoDB replication?
30. How does automatic failover work in a MongoDB replica set?

Sharding and Scalability:

31. Describe sharding in MongoDB and its use for horizontal scalability.
32. How do you enable sharding for a MongoDB collection?
33. What is a shard key, and how do you choose an appropriate one?
34. Explain the difference between "routed" and "scattered" queries in a sharded environment.
35. How does MongoDB ensure data distribution and load balancing in sharded clusters?

Security in MongoDB:

36. What security measures can you implement to protect MongoDB from unauthorized access?
37. How do you enable authentication and authorization in MongoDB?
38. Explain the role of roles and privileges in MongoDB's access control.
39. How can you enable SSL/TLS encryption for MongoDB connections?
40. What is client-side field-level encryption in MongoDB, and when should it be used?

Backup and Restoration:

41. Describe the methods and tools available for creating backups in MongoDB.
42. How do you perform a point-in-time restore in MongoDB?
43. What is the purpose of the "mongodump" and "mongorestore" utilities?
44. Explain the significance of snapshot and journaling in MongoDB backups.
45. How can you automate the backup process in MongoDB?

Monitoring and Performance Optimization:

46. What tools and techniques are available for monitoring MongoDB performance?
47. How do you identify slow queries in MongoDB and optimize them?
48. Describe the purpose of the MongoDB profiler and how to enable it.
49. What is the significance of the "working set" in MongoDB performance optimization?
50. How can you scale MongoDB horizontally and vertically to handle increased loads?